CityInsight: Incorporating Dual-Condition-Based Diffusion Model Into Building Footprint Segmentation From Remote Sensing Imagery

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***Abstract*— Accurately identifying urban building layouts plays a crucial role in understanding the complexity of urban construction and the level of economic development. Previous footprint segmentation methods have struggled to adapt to the diverse morphology of buildings, limiting the accurate extraction of building footprints from remote sensing imagery and impeding insights and understanding of urban areas. To this end, we propose a framework named CityInsight for analyzing urban building morphology from remote sensing imagery. First, we establish a semantic segmentation network, dual-condition diffusion network (DC-Net), based on a diffusion model to accurately identify building footprints from remote sensing images. Second, we use uncertainty attention and condition attention to generate spatial and semantic priors. Finally, we design a condition injection module to incorporate spatial and semantic information into the diffusion learning. Comprehensive experiments demonstrate the accuracy, robustness, and generalization of the proposed method. The F_1-scores of the DC-Net on the large-scale remote sensing datasets SpaceNet, WHU Building, Inria, and Massachusetts are 92.05%, 96.59%, 92.17%, and 92.86%, respectively. Furthermore, the footprint segmentation is utilized for subsequent urban function identification and urban analysis of the Zona Oeste of Rio de Janeiro, to emphasize the application value of CityInsight. Our code is available at https://github.com/Ting-Devin-Han/CityInsight**

***Index Terms*— Building, deep learning, diffusion model, footprint segmentation, remote sensing.**

I. Introduction

EXTRACTING buildings from remote sensing images plays an important role in urban perception [1], [2], understanding [3], planning [4], and construction [5]. Manual delineation of buildings is laborious and time-consuming, which compels researchers to advance automated techniques for building extraction. Unfortunately, as buildings vary in scale, shape, occlusion, and scene, accurately extracting building footprints remains a challenging task.

In recent years, the remote sensing community has shown the potential for high accuracy in building footprint segmentation using deep learning methods, including approaches based on convolutional neural networks (CNNs) [6], Vision Transformers [7], and generative adversarial networks (GANs) [8]. However, these methods are not always sensitive to the variability, irregularity, and density of buildings, making them sometimes ineffective for complex scenarios and potentially compromising the credibility of subsequent urban analyses.

The recent diffusion probabilistic models (DPMs) [9] have garnered widespread attention for their high-quality, diverse generative capabilities, and have demonstrated strong generalization for semantic segmentation under various conditions [10], [11], [12]. Designing a segmentation framework grounded in diffusion models facilitates the representation of multifaceted urban building morphology.

To this end, we aim to provide a novel framework named CityInsight to explore urban morphology from remote sensing images. Instead of focusing on rooftop delineation or geometric reconstruction, our work aims to generate pixel-level building footprint masks suitable for downstream urban spatial analysis. First, a building footprint segmentation method named dual-condition diffusion network (DC-Net) is proposed with improved accuracy and better generalization. It allows for the precise extraction of buildings from remote sensing imagery to analyze urban morphology and layout. Second, we build the uncertainty attention and condition attention to extract edge features and semantic features as priors. Finally, priors are injected into diffusion learning by the condition injection module for robust building footprint segmentation.

Extensive experiments are conducted on the four well-known remote sensing datasets, such as the SpaceNet, WHU Building, Inria Aerial Image, and Massachusetts Building datasets. Moreover, to accommodate different geographical conditions and image resolutions, we mainly use the WHU-Building dataset and combine it with other datasets for training. Even with the hybrid data, the proposed

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method still achieves comparable performance. The main contributions are as follows.

1) The DC-Net is designed to extract the varied, dense, and irregular footprints of buildings in complex urban environments.
2) The dual condition is used to generate spatial and semantic priors, and these priors are injected to optimize the diffusion learning.
3) The proposed method is validated on different datasets to demonstrate the performance of building footprint segmentation.

II. Related Work

A. Handcrafted Feature-Based Methods

Building footprint segmentation is the process of utilizing semantic segmentation methods to extract building masks from remote sensing images, identifying buildings (foreground) from the scene (background). The early methods relied on handcrafted features using digital image processing and computer vision techniques such as co-occurrence matrices [13], contour extraction [14], stochastic geometry [15], and edge gradients [16]. However, handcrafted feature methods struggle to adapt to complex scenes and varying building shapes.

B. CNN-Based Methods

Since the development of deep learning, the self-learning-based neural network approaches have been applied to building footprint segmentation [17], [18]. The end-to-end FCN [19] laid the foundation for deep learning methods in building footprint segmentation [20], inspiring the development of the MA-FCN [21], which utilized fully convolutional networks to extract buildings from remote sensing imagery. The ConvNet [22] introduced a signed distance function in the convolutional network. The UNet [23] is a U-shaped architecture to efficiently capture semantic features. Afterward, the UNet has been widely adopted as a mainstream framework for building footprint segmentation [24], [25]. Holistically nested edge detection [26] was embedded into the UNet backbone to share information between edges and segmentation masks. In particular, the above-mentioned methods ignored the transfer and interaction between different feature information and pixel relationships in a global manner.

Beyond semantic segmentation, convolutional methods have also been applied to structured output prediction and polygonal building delineation. Another significant contribution of the ConvNet [22] is the utilization of a geographic information system (GIS) to generate labeled contour polygon data. The PolyCNN [27] leveraged the FCN to construct a vector extraction network and predicted the vertices of building polygons. Polygonal Building [28] utilized the UNet to extract building segmentation masks and learned to align spatial gradients using frame fields. PolyMapper [29] extracted topological road and building maps from overhead imagery through structured output learning. HiSup [30] showed that the learned reversible building masks take all the merits of the advances of deep CNNs for high-performing polygonal mapping of buildings. Vectorized rooftop delineation methods, such as Polyworld [31], employed graph convolutional networks to locate key points and edges of polygons for building segmentation. Currently, HigherNet-DST [32] has overcome the scale variance problem in polygon-based rooftop delineation.

While these methods are effective in polygonal or vectorized representation, these methods typically focus on rooftop or contour vectorization for fine-grained modeling. In contrast, our work targets pixel-level building footprint segmentation, which better supports urban-scale spatial metrics and widely used remote sensing benchmarks.

C. Transformer-Based Methods

The development of the visual Transformer [33] has overcome the limitations of finite receptive fields in CNNs. By employing attention mechanisms [34] to extract global information, it significantly enhances the robustness of building footprint segmentation. In the process of utilizing the Transformer to capture global relationships, BuildFormer [35] encoded spatial details. UNetFormer [36] constructed a UNet-like architecture and developed an efficient global–local attention mechanism. RSBuilding [37] integrates spatiotemporal clues of building change detection into the Transformer, enhancing the complementarity of different tasks. Sparse Token Transformers [38] provided an efficient dual-pathway Transformer framework to build a sparse attention mechanism, achieving better performance with low time cost.

However, Transformer-based methods still face challenges in terms of generalization across different datasets and scenes, being unable to adapt to diverse scenarios and variable building instances. The Line2Poly [39] combined CNNs and Transformers to predict polygonal topology based on building outlines. However, the polygon-based methods had inherent regularity, making it difficult to adapt to irregular buildings.

D. Generative Models-Based Methods

Previous methods based on CNNs and Transformers overlook the latent relationships between pixels, thus lacking the concept of data distribution. GANs [40] addressed the issue of correlation learning through implicit modeling, fitting the original data distribution to achieve better segmentation. GAN-SCA [41] used a GAN with spatial and channel attention mechanisms to reduce the uncertainty in building segmentation. O-GAN [42] consisted of two orthogonal GANs to learn intraclass variation. MSR-cGAN [35] was proposed based on a multiscale recurrent conditional GAN for building segmentation in dense urban areas.

DPM [9] is a class of generative models that approximate the distribution of image pixels. Their performance surpasses that of GANs in multiple domains [43], [44], especially in image semantic segmentation [10], [11]. On the one hand, compared to GANs, diffusion models offer more stability during training and capture finer details as the process progresses step by step. On the other hand, the introduction of conditions enhances the controllability and interpretability of diffusion models. Recently, Gedara Chaminda Bandara et al. [45] introduced the diffusion model into change detection

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from remote sensing imagery, highlighting the pivotal role of pretrained DPMs as a feature extractor for downstream applications. SegDiff [46] and Diffseg [12] explored the potential of diffusion models for semantic segmentation in remote sensing. This inspired us to apply diffusion models to building footprint segmentation, adapting to the complex urban environment to accommodate the varied, dense, and irregular shapes of buildings.

Compared to CNN- and Transformer-based segmentation networks, diffusion models gradually reconstruct the target through iterative denoising steps, which enables better modeling of uncertain boundaries and ambiguous spatial structures. This generative framework captures both global semantic consistency and local detail preservation, offering a unique advantage in complex urban scenarios where footprints may suffer from occlusion, shadow, or structural complexity. It is worth noting that our method is not designed to produce vectorized rooftop outlines but rather raster-based building footprint masks that support geospatial computation and urban morphology modeling.

III. Method

We proposed a diffusion model-based segmentation architecture for building footprint extraction, named DC-Net. The DC-Net consists of three components: condition extraction, condition injection, and diffusion learning, as shown in Fig. 1. Two CNNs with separate weights are employed for condition extraction and diffusion learning. To effectively embed the extracted condition features into the diffusion process, we design a condition injection module to incorporate spatial and semantic information into the diffusion learning. Specifically, uncertainty attention and condition attention provide edge features and deep semantic features, respectively.

A. Background of the Diffusion Model

DPMs are a type of deep learning-based generative model similar to GANs and variational autoencoders (VAEs). It, abbreviated as the diffusion model, is a parameterized Markov process to generate targets from noise. The diffusion model consists of a forward stage and a reverse stage. The forward process refers to a Markov chain that gradually adds noise to the data, and it is learned to reverse a diffusion process.

Specifically, the diffusion process is formed as $p_\theta(x_0) := \int p_\theta(x_{0:T})\mathrm{dx}_{1:T}$, where $x_1, \ldots, x_T$ are noise to the data $x_0 \sim q(x_0)$. The approximate posterior $q(x_{1:T}|x_0)$ is gradually added Gaussian noise $x_T \sim \mathcal{N}(x_T; 0, I)$ according to a variance schedule $\beta_1, \ldots, \beta_T$, it is mathematically formed as $q(x_{1:T}|x_0) := \prod_{t=1}^{T} q(x_t|x_{t-1})$, where $q(x_t|x_{t-1} := \mathcal{N}(x_t; (1-\beta_t)^{1/2}x_{t-1}, \beta_t I)$. The reverse process is a joint distribution $p_\theta(x_{0:T})$ with learned Gaussian transitions $x_T \sim \mathcal{N}(x_T; 0, I)$, its process is denoted as $p_\theta := p(x_T)\prod_{t=1}^{T} p_\theta(x_{t-1}|x_t)$, where $p_\theta(x_{t-1}|x_t) := \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \sum_\theta(x_t, t))$. Without considering variance, μ_θ can be mathematically formulated as $\mu_\theta = (1/(\alpha_t)^{1/2})(x_t - (1-\alpha_t/(1-\alpha_t)^{1/2})\epsilon_\theta(x_t, t))$. Thus, we can train the reverse process and use a CNN to predict $\epsilon_\theta(x_t, t)$ from x_t. The model $\epsilon_\theta(x_t, t)$ is widely adopted in the U-Net architecture and its variants.

B. Overall Architecture

First, the condition extraction module takes the raw image as input and obtains multiscale features of the image. We use F^I to represent the features extracted by the condition extractor, which include both the encoder features and their corresponding decoder features with channels of $\{64, 128, 256, 512\}$. Second, the diffusion learning module is utilized to recover the segmentation mask from the noisy image x_T during t steps of the diffusion process. The feature dimension of the diffusion learning module is the same as that of the condition extraction module. Finally, the condition injection module is designed to provide contour priors and semantic priors for diffusion learning, where the uncertainty attention extracts edge information from high-resolution features, and the condition attention module extracts semantic information from low-resolution features.

The final segmentation is predicted through the diffusion learning module. Following the DPM [9], the noise prediction loss is defined as

$$\mathcal{L}_{\mathrm{NP}} = \mathbb{E}_{t,x_0,\epsilon}\left[\| \epsilon - \epsilon_\theta\left(\sqrt{\bar{\alpha}_t}x_0 + \sqrt{1-\bar{\alpha}_t}\epsilon, I, t\right) \|^2\right] \quad (1)$$

where I denotes the raw image and t is uniform between 1 and T. Other symbols are explained in the diffusion learning module. Thus, the total loss is represented as

$$\mathcal{L}_{\mathrm{Total}} = \mathcal{L}_{\mathrm{NP}} + \mathcal{L}_{\mathrm{BCE}} + \mathcal{L}_{\mathrm{MSE}} \quad (2)$$

where the binary cross-entropy $\mathcal{L}_{\mathrm{BCE}}$ and mean square error $\mathcal{L}_{\mathrm{MSE}}$ are used to supervise condition extraction.

C. Diffusion Learning

The diffusion model is typically formalized as a forward process and a reverse process. Given building footprint label x_0, we gradually add Gaussian noise $x_T \sim \mathcal{N}(0, I)$ to x_0 through a series of steps T to simulate the forward process

$$q(x_{1:T}|x_0) := \prod_{t=1}^{T} q(x_t|x_{t-1}) \quad (3)$$

where $q(x_{1:T}|x_0)$ denotes the approximate posterior calculated through the variance schedule β and x_t is the noisy sample in the forward process. x_t with added noise is restored to the original data during the reverse process. For this stage, a neural network is trained to simulate the denoising process, as represented by

$$p_\theta(x_{0:T}) := p(x_T)\prod_{t=1}^{T} p_\theta(x_{t-1}|x_t) \quad (4)$$

where $p_\theta(x_0)$ is the probability density function of x_0, $p(x_T)$ represents the variable distribution that follows $\mathcal{N}(x_T; 0, I)$, and $p_\theta(x_{0:T})$ denotes the joint distribution. According to the equation above, given the input segmentation label image $I \in \mathbb{R}^{H\times W}$, the reverse process learns transitions from $p_\theta(x_T)$ to $p_\theta(x_0)$ starting at $p_\theta(x_T) = \mathcal{N}(x_T; 0, I_{H\times W})$. Similar to the forward process, the reverse stage is also completed through a series of steps T, with the final output being the predicted segmentation result.

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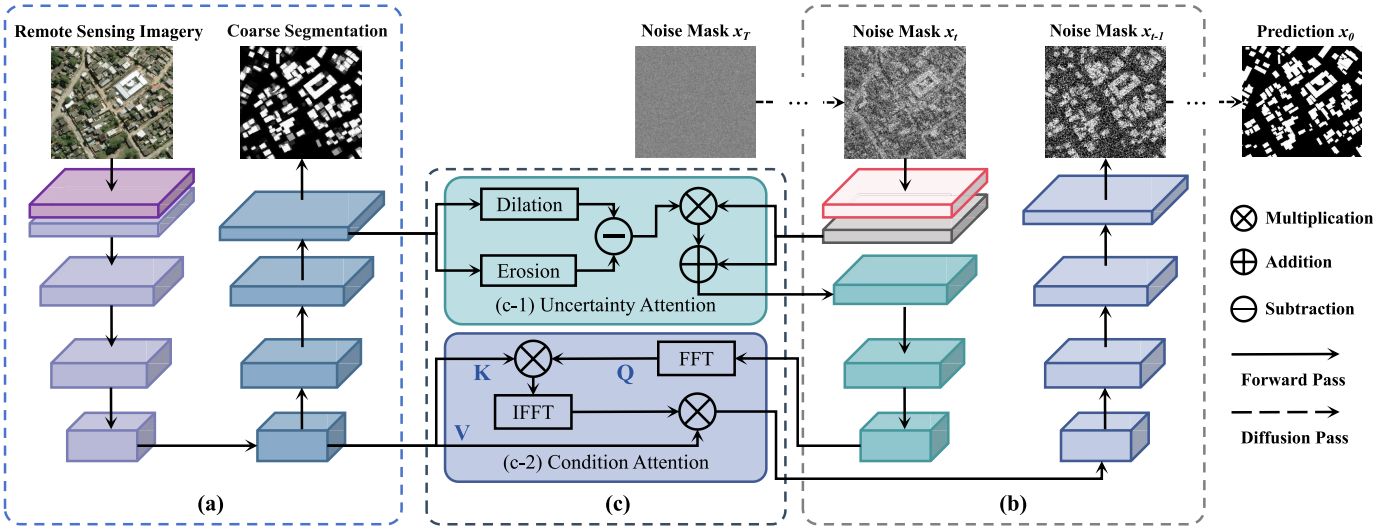


Fig. 1. Overall architecture of the proposed DC-Net, which retains (a) condition extraction, (b) diffusion learning, and (c) condition injection module. The original image is used to extract spatial and semantic priors from CNN-based condition extraction. The prior information is fused into the noisy features through the condition injection module to predict the segmentation mask in diffusion learning, where x_t denotes the current noisy input at timestep t. Condition injection module consists of (c-1) uncertainty attention and (c-2) condition attention, where dilation and erosion denote the morphological operations, and FFT and IFFT represent the fast Fourier transform and inverse FFT, respectively.

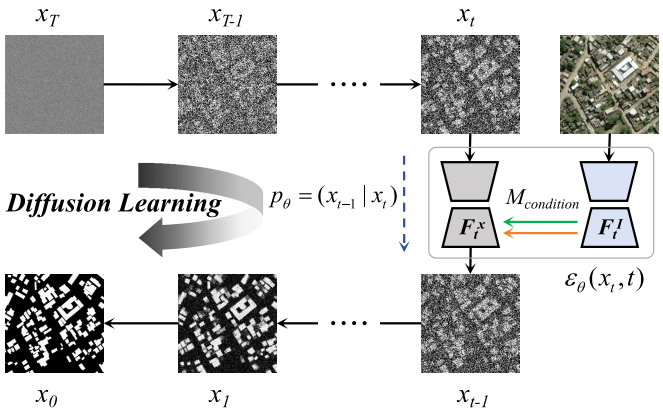


Fig. 2. Pipeline of the diffusion process. A CNN is employed to predict $\epsilon_\theta(x_t, t)$ from x_t, and the other CNN injects the condition to the diffusion learning via $M_{\text{condition}}$.

Following the standard implementation of the DPM, we held the variance schedule β constant as hyperparameters and predicted Gaussian conditionals in $p_\theta(x_{t-1}|x_t)$. Given a real image $I \in \mathbb{R}^{H\times W\times 3}$, as shown in Fig. 2, the CNN is employed to condition noise predict function $\epsilon_\theta(x_t, t)$

$$\epsilon_\theta(x_t, I, t) = \text{DL}\left(M_{\text{Condition}}\left(F_t^I, F_t^x\right), t\right) \tag{5}$$

where F_t^I and F_t^x represent the real image features and noised features in the current step t, respectively, and DL denotes the diffusion learning module. Those two features are incorporated together to fuse and transform via the function $M_{\text{Condition}}$ to inject condition into the reverse process. By this means, the final prediction of label masks for real images is achieved.

D. Condition Injection Module

To introduce prior conditions from the raw image I into diffusion learning, we integrate features from the condition extraction into the segmentation features of the current step x^t. Specifically, uncertainty attention and condition attention are designed to achieve feature-level fusion for injecting conditions.

Uncertainty attention condition provides contour conditions for the diffusion model, enabling the diffusion model to be aware of the positional priors of the mask to optimize segmentation results. Following our previous work [47], we use edge information to construct uncertainty attention. In contrast, due to the difficulty in balancing the uncertainty of grayscale values, we did not utilize the present thresholds for foreground and background information. Traditional image-processing techniques, such as erosion and dilation, are used to compute edge information. As shown in Fig. 1(c-1), we perform dilation and erosion on the output layer feature F^{I_1} of the condition extraction module. Subsequently, the result of dilation minus erosion is multiplied channel-wise with the features F^{x_1} of the diffusion learning module to form channel attention. Channel attention is combined with the features F^{x_1} pixel-wise to obtain the features enhanced by uncertainty attention, injecting spatial priors into diffusion learning in a manner optimized through edge information. We define F^{x_2} as the output feature of the uncertainty attention. The computational process of the uncertainty attention module is mathematically formalized as follows:

$$F^{x_2} = \left[\left(\text{Dil}\left(F^{I_1}\right) - \text{Ero}\left(F^{I_1}\right)\right) \times F^{x_1}\right] + F^{x_1}. \tag{6}$$

Inspired by classical morphological operations, we simulate dilation and erosion using depthwise separable convolutions with learnable kernels 2×2. To adaptively combine the outputs of dilation and erosion branches, we introduce learnable confidence maps and normalize minus via a Softmax function.

The intermediate layer features of the condition extraction module contain rich semantic information, which can provide semantic priors for diffusion learning to optimize feature extraction. Therefore, inspired by MedSegDiff [11], condition attention is designed to embed semantic priors into diffusion learning with a cross-attention architecture, as shown

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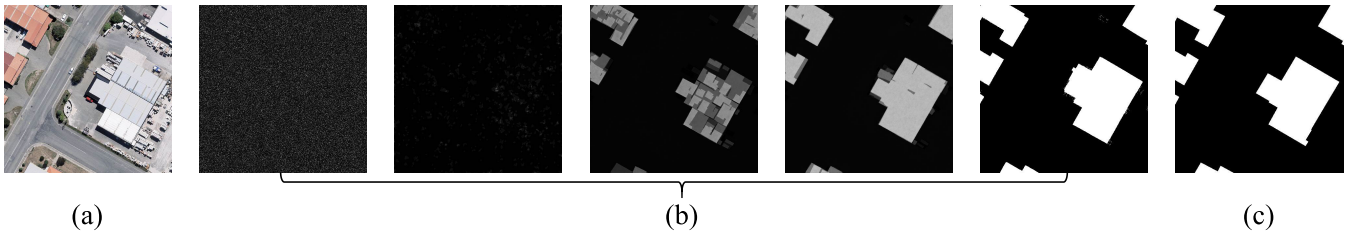


Fig. 3. Diffusion sampling and ensemble result, where (a) is the original input, (b) represents the sampling process and ensemble result, and (c) is the GT.

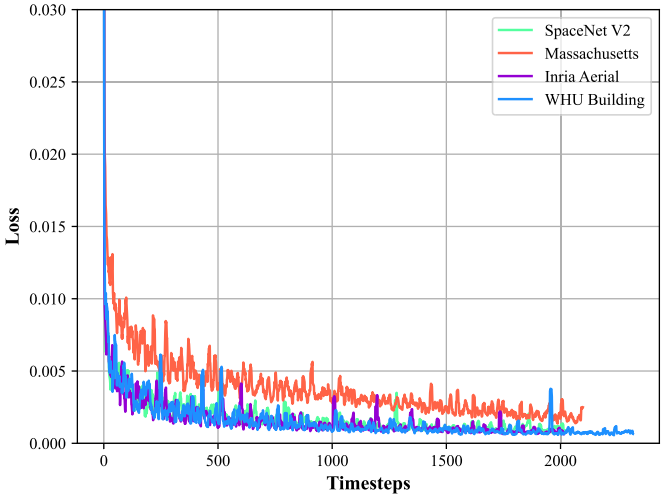


Fig. 4. Training process of the proposed DC-Net among four datasets.

in Fig. 1(c-2). However, using cross-attention to fuse F^{I_4} and F^{x_4} introduces redundant high-frequency noise, increasing the difficulty of convergence. Two-dimensional fast Fourier transform (FFT) $\mathcal{F}(\cdot)$ is used to constrain the noise-related components in F^x. First, F^{x_4} and F^{I_4} are transformed into the query, key, and value for cross-attention. The query generated from F^{x_4} is performed via 2-D FFT along the spatial dimensions. Second, following the form of cross-attention, the query and key are multiplied together to build the attention map and then inverse FFT (IFFT) $\mathcal{F}^{-1}(\cdot)$ transformed it back to the original spatial domain. Finally, the attention map and value are pixel-wise multiplied to generate diffusion learning features with semantic priors. The formulation of conditional attention is as follows:

$$F^{x_{4'}} = \left[\mathcal{F}^{-1}\left(\mathcal{F}\left(F^{x_4}\right) \cdot F^{I_4}\right)\right] \cdot F^{I_4}. \tag{7}$$

The condition injection module, incorporating uncertainty attention and condition attention, is considered to provide spatial priors and semantic priors for diffusion learning in a learnable manner. On the one hand, it constrains the regions of interest of diffusion features through edge information. On the other hand, based on constrained frequencies, it establishes global interactions between condition features and diffusion features at the semantic level. The proposed module is able to be embedded into various frameworks to enhance features.

IV. Experiments

A. Datasets

1) SpaceNet: The SpaceNet dataset challenges aim to extract building footprints from satellite imagery, including versions 1 and 2. V1 encompasses 2544 km^2 of Rio de Janeiro with 137 743 buildings labeled. V2 covers Las Vegas, Paris, Shanghai, and Khartoum, and it includes 302 701 labeled buildings. The dataset is divided into training, validation, and test data in a 6:2:2 ratio, with each image having a resolution of 650 × 650 pixels.

2) WHU Building: The aerial data in the WHU Building dataset consists of more than 220 000 buildings and 450 km^2 covering in Christchurch, New Zealand. The size of each aerial image is 512 × 512 pixels.

3) Inria Aerial Image: The Inria Aerial Image Labeling dataset covers 810 km^2 and includes ten different densely populated or mountainous urban areas, such as Chicago, San Francisco, and Lienz. The original images in the dataset have a resolution of 5000 × 5000 pixels, which are cropped to a resolution of 500 × 500 for training and testing purposes.

4) Massachusetts Building: The Massachusetts dataset covers a 340 km^2 area of building coverage in the Boston region. The original image pixels are 1500 × 1500, which are cropped to 500 × 500.

B. Metrics

We consider building footprint segmentation as a binary classification task, and we use intersection over union (IoU), recall (REC), precision (PRE), and F_1-score (F_1) as evaluation metrics, which are formulated as follows:

$$\begin{aligned} \text{IoU} &= \frac{\text{TP}}{\text{FN}+\text{TP}+\text{FP}} \\ \text{REC} &= \frac{\text{TP}}{\text{TP}+\text{FN}} \\ \text{PRE} &= \frac{\text{TP}}{\text{TP}+\text{FP}} \\ F1 &= 2 \times \frac{\text{REC} \times \text{PRE}}{\text{REC}+\text{PRE}} \end{aligned} \tag{8}$$

where TP, FP, TN, and FN denote the true positive, false positive, true negative, and false negative in the confusion matrix, respectively.

C. Implementation Details

The experiments of building footprint segmentation are conducted using the Linux Ubuntu platform with CUDA 11.7 and PyTorch 2.2.1. All the models are trained and tested on a single NVIDIA A100 80G GPU. The models are trained with a batch size of 4. The initial learning rate is set to 1e^{-}4 with the AdamW optimizer. The diffusion step is set to 1000 for the inference, and the models are set ten times for the ensemble. Once the loss has decreased to a certain extent, it will not

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TABLE I
COMPREHENSIVE EVALUATION RESULTS OF DIFFERENT METHODS ON THE FOUR BUILDING FOOTPRINT SEGMENTATION DATASET. BEST RESULTS ARE DENOTED AS **BOLD**

Method	Type	SpaceNet	WHU		Inria		Massachusetts	
		F_1 (%)	IoU (%)	F_1 (%)	IoU (%)	F_1 (%)	IoU (%)	F_1 (%)
U-Net [23]	CNN	70.04	89.60	94.52	76.44	86.65	67.61	80.68
MA-FCN [21]	CNN	-	89.90	94.68	79.67	88.68	73.80	84.93
TreeUNet [48]	CNN	-	-	-	76.52	81.46	74.13	78.26
BRRNet [49]	CNN	-	-	-	82.46	89.13	79.36	86.92
Bias-UNet [50]	CNN	-	-	-	-	-	73.49	84.72
CBR-Net [51]	CNN	49.06	91.40	95.51	81.10	89.56	74.55	85.42
HRNet [52]	CNN	67.82	90.00	94.74	80.03	88.91	72.22	83.87
BRNet [53]	CNN	89.60	-	-	-	-	-	-
BSPNet [54]	CNN	72.98	-	-	-	-	-	-
Easy-Net [2]	CNN	-	89.16	94.20	-	-	-	-
FD-Net [55]	CNN	-	91.14	95.36	-	-	74.54	85.42
UANet [56]	CNN	-	92.15	95.91	83.08	90.76	76.41	86.63
HD-Net [6]	CNN	-	90.19	94.84	82.10	90.17	75.53	86.06
MANet [57]	Transformer	-	-	-	-	-	70.76	82.88
SegFormer [33]	Transformer	-	89.70	94.57	78.59	88.01	-	-
MAP-Net [58]	Transformer	-	90.86	95.21	-	-	-	-
BuildFormer [35]	Transformer	-	91.44	95.53	81.44	89.77	75.74	86.19
UniFormer [59]	Transformer	-	90.55	95.04	84.37	91.52	73.80	84.92
BCTNet [60]	Transformer	-	91.15	95.37	-	-	75.04	85.74
DPENet [61]	Transformer	-	-	-	-	-	76.69	86.81
CLGFF-Net [62]	Transformer	-	91.30	95.45	84.94	91.85	75.33	85.93
RSBuildig [37]	Transformer	-	92.15	95.88	82.68	90.52	-	-
MSR-cGAN [8]	Generative	-	-	-	**86.15**	**92.47**	85.41	91.27
DiffSeg [12]	Generative	-	-	-	-	-	67.11	80.03
DC-Net Trained on Hybrid (Ours)	Generative	86.62	-	-	81.80	89.76	77.16	85.15
DC-Net (Ours)	Generative	**92.05**	**93.41**	**96.59**	85.56	92.17	**87.17**	**92.86**

continue to decrease, but the segmentation performance will still improve.

Fig. 3 shows predictions at different steps and the ensemble result. The remote sensing images are resized to a resolution of 512 × 512 pixels. We trained different models for different datasets; however, to validate the generalization ability of the methods, we used the model trained on the hybrid dataset as the base model to test the images from other datasets. The change of losses during the training process is shown in Fig. 4. The model demonstrates a rapid decline in loss in the first 200 iterations for all datasets, indicating fast convergence. Among them, the Inria and WHU datasets show smoother and more stable optimization curves, while the Massachusetts dataset exhibits relatively higher initial loss and greater fluctuation, due to the complex urban forms and dense buildings. Overall, the consistent downward trend across all datasets validates the generalization and training stability of the DC-Net.

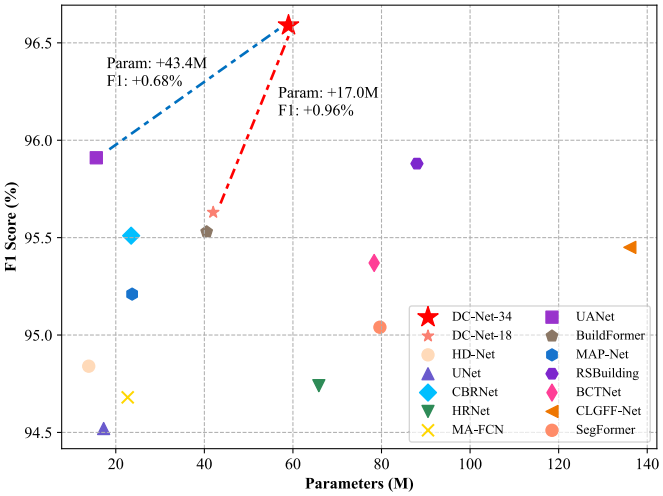


Fig. 5. F_1-score versus parameters of different methods on the WHU building dataset.

D. Quantitative Evaluation

Table I presents the quantitative comparison results between the proposed DC-Net and other SOTA methods on four well-known large-scale remote sensing datasets. In the showcased metrics of IoU and $F1$, the DC-Net achieves the highest building segmentation accuracy in the SpaceNet-V2, WHU Building, and Massachusetts datasets, and the second-best in the Inria Aerial Image dataset. Specifically, the DC-Net demonstrates a 2.45% improvement over the previously best-performing BRNet. Meanwhile, the DC-Net surpasses both metrics of the latest series of methods in the WHU Building dataset.

The performance gain can be attributed to the iterative generation mechanism of diffusion models, which progressively denoise semantic representations and enable more accurate boundary delineation. Unlike CNNs and Transformer-based methods, which rely on one-pass prediction, our network refines structural predictions step by step, making them inherently more robust to local ambiguities and spatial inconsistencies.

As shown in Fig. 5, the DC-Net achieved the highest accuracy, with a 0.68% gain over the second-best method. However, this improvement came at the cost of an additional 43.4 M parameters. Compared to selecting a lighter backbone

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TABLE II

COMPREHENSIVE EVALUATION RESULTS OF DIFFERENT METHODS ON THE SPACENET V2 DATASET. BEST RESULTS ARE DENOTED AS **BOLD**

Method	REC (%)	PRE (%)	F_1 (%)
U-Net [23]	88.79	62.47	70.04
CBR-Net [51]	48.39	65.81	49.06
HRNet [52]	73.18	72.01	67.82
BRNet [53]	88.80	91.20	89.60
BSPNet [54]	85.16	69.42	72.98
DC-Net (Ours)	**91.79** (+2.99)	**92.57** (+1.37)	**92.05** (+2.45)

TABLE III

COMPREHENSIVE EVALUATION RESULTS OF DIFFERENT METHODS ON THE WHU BUILDING DATASET. BEST RESULTS ARE DENOTED AS **BOLD**

Method	IoU (%)	REC (%)	PRE (%)	F_1 (%)
U-Net [23]	89.60	94.77	94.52	94.52
SegFormer [33]	89.70	94.42	94.72	94.57
MAP-Net [58]	90.86	94.81	95.62	95.21
CBR-Net [51]	91.40	95.70	95.31	95.51
HRNet [52]	90.00	94.34	95.14	94.74
Easy-Net [2]	89.16	93.48	94.94	94.20
BuildFormer [35]	91.44	95.65	95.40	95.53
UniFormer [59]	90.55	95.07	95.01	95.04
BCTNet [60]	91.15	95.27	95.47	95.37
FD-Net [55]	91.14	95.46	95.27	95.36
UANet [56]	92.15	95.86	95.96	95.91
HD-Net [6]	90.19	94.68	95.00	94.84
CLGFF-Net [62]	91.30	95.89	95.01	95.45
RSBuildig [37]	92.15	95.82	95.93	95.88
DC-Net (Ours)	**93.41** (+1.26)	**97.23** (+1.34)	**96.01** (+0.05)	**96.59** (+0.68)

network, using ResNet-34 as the backbone provided sufficient improvement over ResNet-18. It is worth noting that we compared two generative model-based segmentation algorithms, MSR-cGAN and DiffSeg, in the Inria and Massachusetts datasets. Compared to DiffSeg, the DC-Net demonstrates a significant improvement of +20.06% and +12.83% in IoU and $F1$ metrics, respectively. This is attributed to the powerful semantic and contour priors provided by the condition attention and uncertainty attention adopted by the DC-Net for the diffusion model.

Compared to the MSR-cGAN, the proposed DC-Net gets slightly lower IoU and $F1$-scores in Inria (−0.59%/−0.30%) but higher IoU and $F1$-scores in Massachusetts (+1.96%/+1.59%). The MSR-cGAN similarly adopts the strategy of injecting semantic priors into the generative model to enhance segmentation performance. Unlike the DC-Net, the MSR-cGAN incorporates a skip connection into the generative model to provide more stable feature representations. To validate the generalization capability of the DC-Net, we used the model trained on the Hybrid dataset to validate on other datasets. As shown in Table I, the proposed method achieves comparable levels to the latest methods on all three datasets. In summary, the DC-Net attains competitive performance on the above four datasets.

Tables II–V present the quantitative evaluation results of our proposed DC-Net compared to SOTA methods on four datasets, with the best results highlighted in bold. By comparing the results, it is evident that the DC-Net demonstrates competitive performance, surpassing recent studies in nearly all metrics. Particularly, the RECs of the DC-Net achieve

TABLE IV

COMPREHENSIVE EVALUATION RESULTS OF DIFFERENT METHODS ON THE INRIA AERIAL IMAGE DATASET. BEST RESULTS ARE DENOTED AS **BOLD**

Method	IoU (%)	REC (%)	PRE (%)	F_1 (%)
U-Net [23]	76.44	86.09	87.21	86.65
TreeUNet [48]	76.52	79.14	85.74	81.46
BRRNet [49]	82.46	86.43	92.02	89.13
SegFormer [33]	78.59	87.08	88.97	88.01
CBR-Net [51]	81.10	89.20	89.93	89.56
HRNet [52]	80.03	87.53	90.33	88.91
BuildFormer [35]	81.44	90.75	88.81	89.77
MSR-cGAN [8]	**86.15**	89.72	**95.41**	**92.47**
UniFormer [59]	84.37	91.21	91.84	91.52
UANet [56]	83.08	89.52	92.04	90.76
HD-Net [6]	82.10	89.26	91.10	90.17
CLGFF-Net [62]	84.94	92.12	91.59	91.85
RSBuildig [37]	82.68	89.65	91.40	90.52
DC-Net (Ours)	85.56 (-0.59)	**96.67** (+6.95)	88.11 (-7.30)	92.17 (-0.30)

TABLE V

COMPREHENSIVE EVALUATION RESULTS OF DIFFERENT METHODS ON THE MASSACHUSETTS BUILDING DATASET. BEST RESULTS ARE DENOTED AS **BOLD**

Method	IoU (%)	REC (%)	PRE (%)	F_1 (%)
U-Net [23]	67.61	82.29	79.13	80.68
MA-FCN [21]	73.80	82.89	87.07	84.93
TreeUNet [48]	74.13	77.52	83.15	78.26
BRRNet [49]	79.36	84.17	89.72	86.92
Bias-UNet [50]	73.49	86.15	83.34	84.72
MANet [57]	70.76	83.77	82.00	82.88
CBR-Net [51]	74.55	84.36	86.50	85.42
HRNet [52]	72.22	82.01	85.82	83.87
BuildFormer [35]	75.74	87.52	84.90	86.19
MSR-cGAN [8]	85.41	88.14	**94.27**	91.27
UniFormer [59]	73.80	82.41	87.60	84.92
DiffSeg [12]	67.11	83.50	77.72	80.03
BCTNet [60]	75.04	83.99	87.57	85.74
FD-Net [55]	74.54	83.02	87.95	85.42
DPENet [61]	76.69	86.77	86.84	86.81
UANet [56]	76.41	85.35	87.94	86.63
HD-Net [6]	75.53	86.13	85.98	86.06
CLGFF-Net [62]	75.33	86.85	85.03	85.93
DC-Net (Ours)	**87.17** (+1.76)	**94.69** (+6.55)	91.22 (-3.05)	**92.86** (+1.59)

top-1 on all four datasets, indicating that the DC-Net is sensitive to backgrounds and is able to predict buildings more stably. However, in the case of PREs on the Inria Aerial and Massachusetts datasets being lower than the MSR-cGAN, it suggests that the rough edge contours of the DC-Net, to some extent, affect the segmentation accuracy. Overall, the DC-Net achieves significant improvement in building footprint segmentation by incorporating semantic and contour priors into the diffusion model, enabling richer feature interactions and leveraging the underlying correlations among different features.

E. Qualitative Evaluation

To give a more intuitive analysis of the DC-Net, the building footprint segmentation results of the DC-Net and several SOTA methods on the WHU Building dataset are provided in Fig. 6. First, sharp boundaries and complete buildings are extracted by the proposed DC-Net. Second, the DC-Net demonstrates stronger adaptability to buildings of various sizes and shapes. Finally, higher accuracy of the DC-Net across different scenarios is exhibited in Fig. 6. Surprisingly, embedding semantic and contour priors into the diffusion model reinforced the accuracy in challenging visual interpretation scenarios. The

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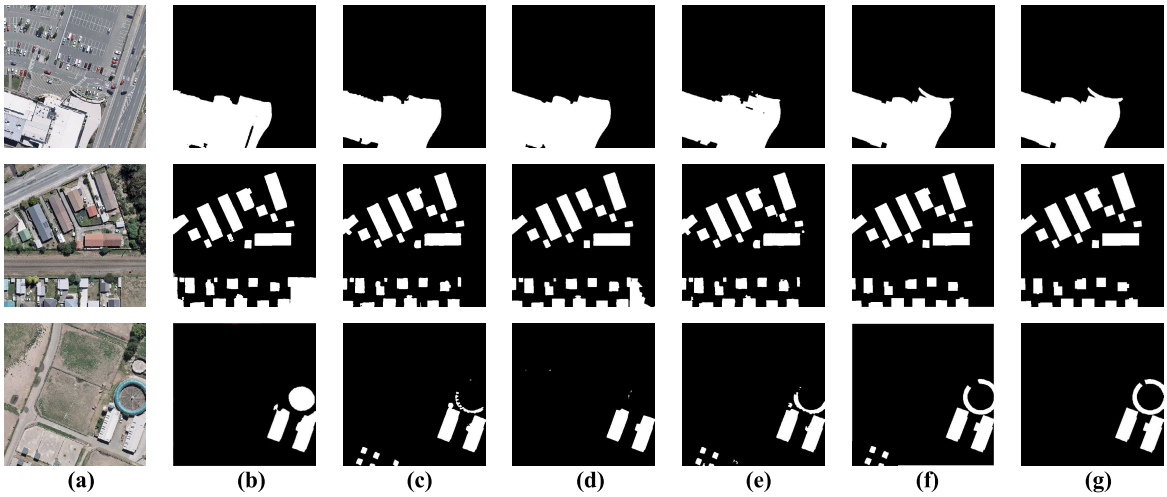


Fig. 6. Building segmentation results in comparison between the proposed DC-Net and other comparative methods on the WHU Building dataset, where (a) is raw images, (b)–(f) represent the predictions of the U-Net, SegFormer, BuildFormer, UANet, our DC-Net, and (g) is the ground truth.

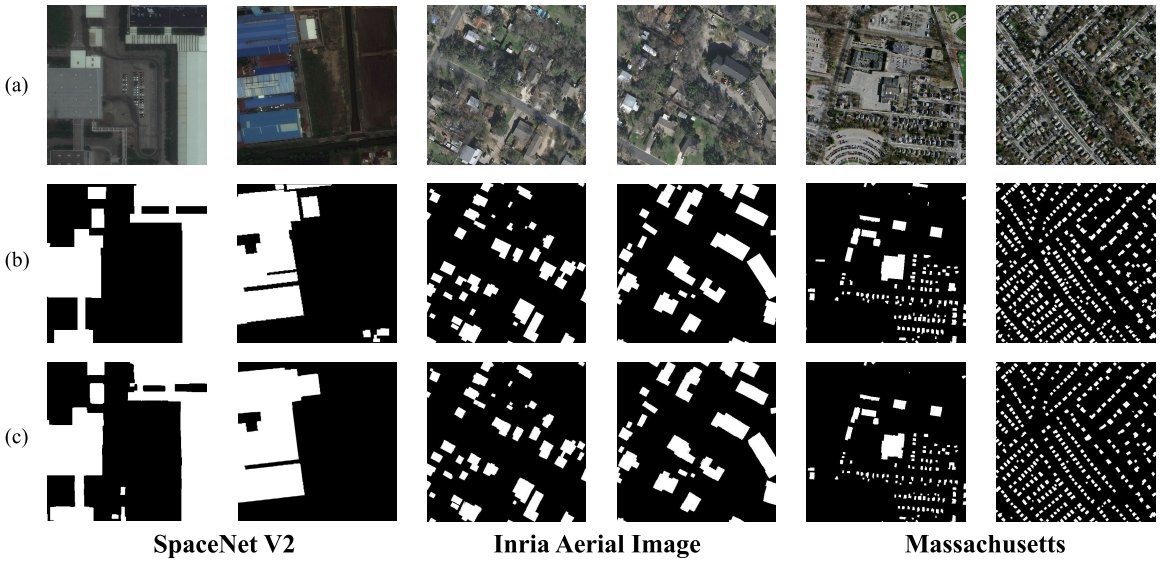


Fig. 7. Building segmentation results of the proposed DC-Net on the SpaceNet V2, Inria Aerial Image, and Massachusetts datasets, where (a) is raw images, (b) represents the ground truth, and (c) is the predictions of our DC-Net.

groundwork for subsequent tasks is laid by the DC-Net, such as building quantity statistics, morphological analysis, and so on.

Compared to CNN and Transformer-based baselines, the DC-Net demonstrates superior stability across datasets and particularly excels in cases with noisy background, adjacent buildings, or incomplete contours. These advantages can be attributed to the iterative denoising nature of diffusion models, which gradually refine predictions and better handle uncertainty.

Fig. 7 presents the visualization results of the proposed DC-Net on the SpaceNet V2, Inria Aerial Image, and Massachusetts datasets. The visualization results reflect the remarkable segmentation capability of the DC-Net, which is able to adapt to various buildings of different sizes, densities, and shapes, as well as different resolutions of remote sensing images. Particularly, even in cases of weak visual clues, buildings obscured by trees, or varying building density (BD), the DC-Net demonstrates the ability to accurately extract building footprints, especially showing robustness against buildings with similar colors or adhering to the background. Furthermore, closely connected instances of different buildings are also distinguished. Although the extracted building boundaries may not be perfectly smooth, the footprint segmentation does not affect subsequent building statistical analysis tasks.

F. Ablation Study

To demonstrate the effectiveness and contributions of different modules in the DC-Net, we conduct ablation experiments on the WHU Building dataset.

During the experiments, we first assessed the effects of modules outlined in the Condition Injection Module, as shown in Table VI. Uncertainty attention and condition attention significantly improve performance over the baseline method. Compared to the baseline, uncertainty attention and condition attention, respectively, increase IoU and $F1$-score by

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TABLE VI
CONTROL ABLATION STUDY RESULTS OF DIFFERENT MODULES IN THE WHU BUILDING DATASET. BEST RESULTS ARE DENOTED AS **BOLD**

Baseline		IoU(%)	F_1 (%)
U-Net		89.60	94.52
Uncertainty Attention	Condition Attention		
		90.35	94.92
✓		91.30	95.44
	✓	91.94	95.75
✓	✓	**93.41**	**96.59**

TABLE VII
CONTROL ABLATION STUDY RESULTS OF DIFFERENT BACKBONES IN THE WHU BUILDING DATASET. BEST RESULTS ARE DENOTED AS **BOLD**

Backbone	Params (M)	IoU (%)	F_1 (%)
ResNet-18	42	91.73	95.63
ResNet-34	59	93.41	**96.59**
ResNet-50	123	**93.46**	96.57
ResNet-101	192	93.33	96.50

0.95%/1.59% and 0.52%/0.83%. This highlights that incorporating richer semantic priors and contour priors into the diffusion model is able to effectively improve the performance of building footprint segmentation. In fact, comparing the performance of the two modules, it was observed that semantic priors yielded greater benefits than contour priors. Therefore, it is necessary to develop more reasonable contour prior algorithms to enhance the perception of building edges and smooth rough edges. Building upon contour priors, the inclusion of semantic information led to a significant performance improvement. This is attributed to the synergistic optimization of both by integrating the relevance of different features, enhancing the ability of the DC-Net to capture building main bodies and boundaries.

To explore the contribution of semantic information with condition attention for building footprint segmentation, we select backbones with different deep layers for experimentation. Experimental data is integrated in Table VII. Obviously, as the network depth increases, performance improves, but it also comes with a skyrocketing number of parameters. Networks with deep structures provide rich features, where both semantics and details are adequately expressed. When ResNet-101 is chosen, network degradation occurs in IoU and $F1$-score. Another issue is the accumulation of parameters; despite a slight improvement in IoU, the number of parameters of ResNet-50 is twice that of ResNet-34. After a comprehensive evaluation, as shown in Fig. 8, ResNet-34 is selected as the backbone network for semantic feature extraction to achieve a tradeoff between accuracy and parameter count.

V. LIMITATIONS

A. Coarse Building Footprint Contours

While the DC-Net demonstrates robustness in building footprint extraction, the contours of the footprints remain somewhat rough due to the inherent uncertainty in the prior information. This issue is particularly evident in the segmentation of large buildings. Specifically, the performance of the diffusion model is highly dependent on the quality of the priors, with clear and accurate priors being essential for optimal segmentation results. In future work, we will focus on developing more stable and refined priors to enhance the accuracy and competitiveness of building footprint extraction.

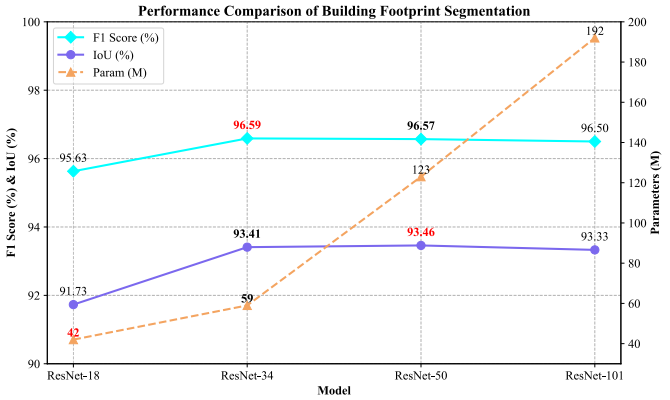


Fig. 8. Visualization compares the F_1-score, IoU, and parameters among different models, where the best performance is marked in red, and the second-best is highlighted in black.

B. Ineffective Utilization of Intermediate Features in the Diffusion Process

Recent diffusion model approaches have begun utilizing temporal features to enhance semantic segmentation. Our observations indicate that, after a sufficient number of iterations, the segmentation results begin to closely approximate accurate building shapes. Based on this, we hypothesize that the intermediate features at various stages of the process could also serve as valuable semantic priors, contributing to improved segmentation performance.

C. High Training Time and Computational Cost

As is well known, one of the key challenges associated with diffusion models is their substantial computational cost and extended training time. Similarly, our proposed method demands a prolonged training phase, incurring high computational overhead. Additionally, the inference process is notably slower compared to traditional deep learning models. In future work, we aim to explore optimization strategies to mitigate the relatively high training and inference costs of diffusion-based segmentation models. Potential directions include knowledge distillation to reduce the number of inference steps, the application of accelerated sampling methods (e.g., denoising diffusion implicit models or stochastic differential equations solvers), and the design of lightweight conditional modules for faster conditioning. These efforts aim to improve computational efficiency while preserving segmentation accuracy.

VI. CONCLUSION

In this paper, we propose a novel framework named DC-Net for urban building segmentation based on a diffusion model from remote sensing images. The DC-Net consists of condition

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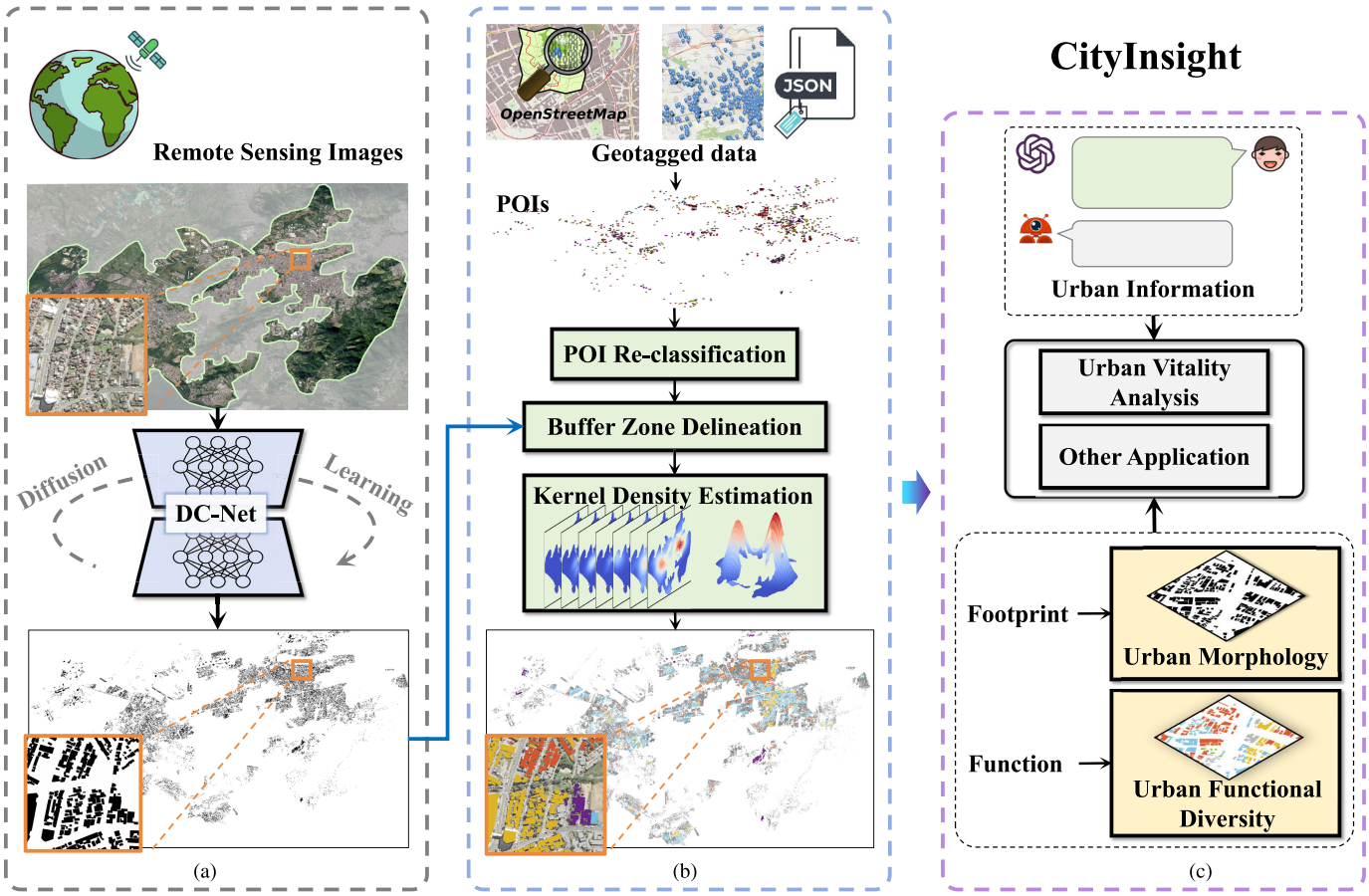


Fig. 9. Overall framework of CityInsight. We separate the CityInsight into (a) DC-Net for building footprint segmentation, (b) urban function identification through the fusion of POIs and building footprints, and (c) urban vitality analysis leveraging large AI models.

extraction, condition injection, and diffusion learning modules. In the condition injection module, we design the edge prior and semantic prior from uncertainty attention and condition attention to provide robust guidance for the diffusion learning. The proposed DC-Net demonstrates outstanding performance in four large-scale remote sensing datasets, surpassing existing state-of-the-art methods and achieving high precision, robustness, and generalized building segmentation.

APPENDIX
APPLICATIONS FOR URBAN VITALITY ANALYSIS

We propose a novel pipeline named CityInsight for identifying building morphology and functionality from remote sensing imagery to analyze urban vitality. We inject POI geographical data into CityInsight to recognize urban building functions within building footprints. The building footprint provides data on urban morphology, while function recognition provides the prerequisite for analyzing urban functions. Urban morphology and functional diversity indicators are calculated to promote the urban data acquisition and urban vitality analysis. The overall framework of CityInsight is illustrated in Fig. 9.

A. Urban Building Function Identification Method

We first categorized the original POI data into various land-use types based on different functional attributes. Subsequent urban analysis is able to be facilitated by this categorization. Following this, we generated a 5-m radius circular buffer [63] for each POI to determine the land use of the encompassed or intersecting building features based on the nearest POI category. Significantly, when a building lies in multiple buffers, its category is assigned based on the POI with the largest overlapping buffer area.

To classify buildings beyond the buffer zone's reach, we employ the KDE method [64], [65] on the POI data, transforming discrete POI points into a continuous density surface indicative of spatial distribution features. The density surface is utilized to determine the functions of buildings. Monitoring the variations in spatial density distribution across diverse POI categories enables us to discern the distribution patterns pertinent to specific functional areas of buildings.

TABLE VIII
SATELLITE IMAGERY DETAILS OF TWO STUDY AREAS

City	Area (km^2)	# Buildings	GSD (m)	Sensor
Rio	2,544	137,743	0.5	WorldView-2
Shanghai	1,000	68,044	0.5	WorldView-3

B. Urban Vitality Analysis Method

1) Urban Morphology: Building indicators are considered the foundation of morphometric studies. The mean building area (MBA), mean building height (MBH), BD,

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TABLE IX
CORRESPONDENCE BETWEEN POI LABELS CATEGORIES, NUMBER OF BUILDINGS, AND NATIONAL STANDARD

Standard Categories	Rio de Janeiro		Shanghai		POI Labels
	# POI	# Building	# POI	# Building	
Residential(R)	774	95,684	3,305	58,749	Residential, Service facilities
Administration and Public Services (A)	425	30,256	1,273	3,686	Administration institutions, Cultural facilities, Education and research, Sports facility, Medical and health care, Social welfare facilities, Cultural relics and monuments, Foreign affairs, Religious facilities
Commercial and Business Facilities (B)	194	4,645	2,789	2,692	Commercial facilities, Business facilities, Entertainment and sports, Public utilities business outlet, Other service facilities
Industrial (M)	129	395	127	213	Factory
Logistics and Warehouse (W)	10	2	423	1,719	Logistics and Warehouse
Street and Transportation (S)	188	96	121	138	Urban road, Urban rail transit, Integrated transportation hub, transport terminal, Other transportation facilities
Municipal Utilities (U)	249	6,665	170	847	Supply facilities, Environment facilities, Safety facilities, Other public facilities

building space index (BSI), and building structure layout index (BSLI) are selected as metrics for urban morphology analysis, where MBH provides a 3-D perspective and the other indicators reflect the 2-D aspect of the city. The higher the MBA, BD, and BSI are, the greater the proportion of built-up areas, indicating a higher level of urbanization. The BSLI is used to describe the spatial layout characteristics of buildings in both horizontal and vertical directions. Mathematical expressions of building indicators are as follows:

$$\begin{aligned} \text{MBA} &= \text{Area}_{\text{Building}}/N_{\text{Building}} \\ \text{BD} &= N_{\text{Building}}/\text{Area}_{\text{City}} \\ \text{BSI} &= \text{Area}_{\text{Building}}/\text{Area}_{\text{City}} \\ \text{BSLI} &= \text{MBA}/\text{MBH} \end{aligned} \tag{9}$$

where $\text{Area}_{\text{Building}}$, N_{Building}, and $\text{Area}_{\text{City}}$ denote the total area of the buildings, the number of buildings, and the area of the city, respectively.

2) Urban Function Diversity: We calculate the function mixture indicator (FXI) to represent the diversity of urban functions, which estimates the composition of functions based on the floor area of each function. Using the results of urban building function identification, we obtain the functional mixture measure through Shannon entropy [66]. The FXI can be formulated as

$$\text{FXI} = -\sum_{i=1}^{n} p_i \ln p_i \tag{10}$$

where p_i denotes the proportion of the ith functional type and n is the number of functional categories. The higher the value of FXI, the greater the diversity of urban building functions, and vice versa. The degree of functional mixture contributes to a more comprehensive understanding of urban vitality and actively promotes urban vitality.

3) Urban Vitality: We integrate a large AI model to further obtain more detailed data and conduct more comprehensive analysis. We utilize the pandas AI toolkit,[1] which combines Pandas and GPT-4,[2] to prompt queries using natural language for obtaining additional urban data. First, we organize and summarize the calculated indicators and then put them into the local database according to the data frame in Pandas. Then, we instantiate the GPT-4 natural language model and access the corresponding data through the data frame and chained prompts by Pandas AI. As urban vitality is closely related to socio-economic indicators, we subsequently utilize GPT-4 to retrieve population, geographic, economic, social, and political factors of the study area. Finally, we analyze the urban vitality of the study area by integrating urban morphology, functional diversity, and socio-economic factors. For comparing multiple cities, we considered indicators at the city level. For individual cities, the urban area was divided into multiple segments to separately calculate different indicators for vitality analysis and comparison.

C. Data and Study Area

Remote sensing images from the SpaceNet dataset of Rio de Janeiro and Shanghai are used for urban function recognition, and then the urban function of Rio de Janeiro is used for urban vitality analysis. The two cities are selected as the research subjects due to certain differences in urban functions, building layout, population density, and public service facilities.

Shanghai is located in the eastern coastal region of China. It is situated approximately at 121°28′25″E and 31°13′49″N. As one of China's major economic, financial, trade, shipping, and technology innovation centers, Shanghai is one of the largest cities in China and one of the most populous cities in the world. Note that the data for the Shanghai area collected from SpaceNet V2 is from the suburban areas of the Pudong New District.

Rio de Janeiro is located in the southeastern part of Brazil, adjacent to the Atlantic Ocean. Its geographical coordinates are approximately 22°54′W and 43°12′N. Specifically, the city of Zona Oeste in Rio de Janeiro is utilized for urban analysis.

Satellite Imagery: The remote sensing images are provided by the SpaceNet dataset; its details are shown in Table VIII.

[1]https://docs.pandas-ai.com/en/latest/

[2]https://openai.com/chatgpt/

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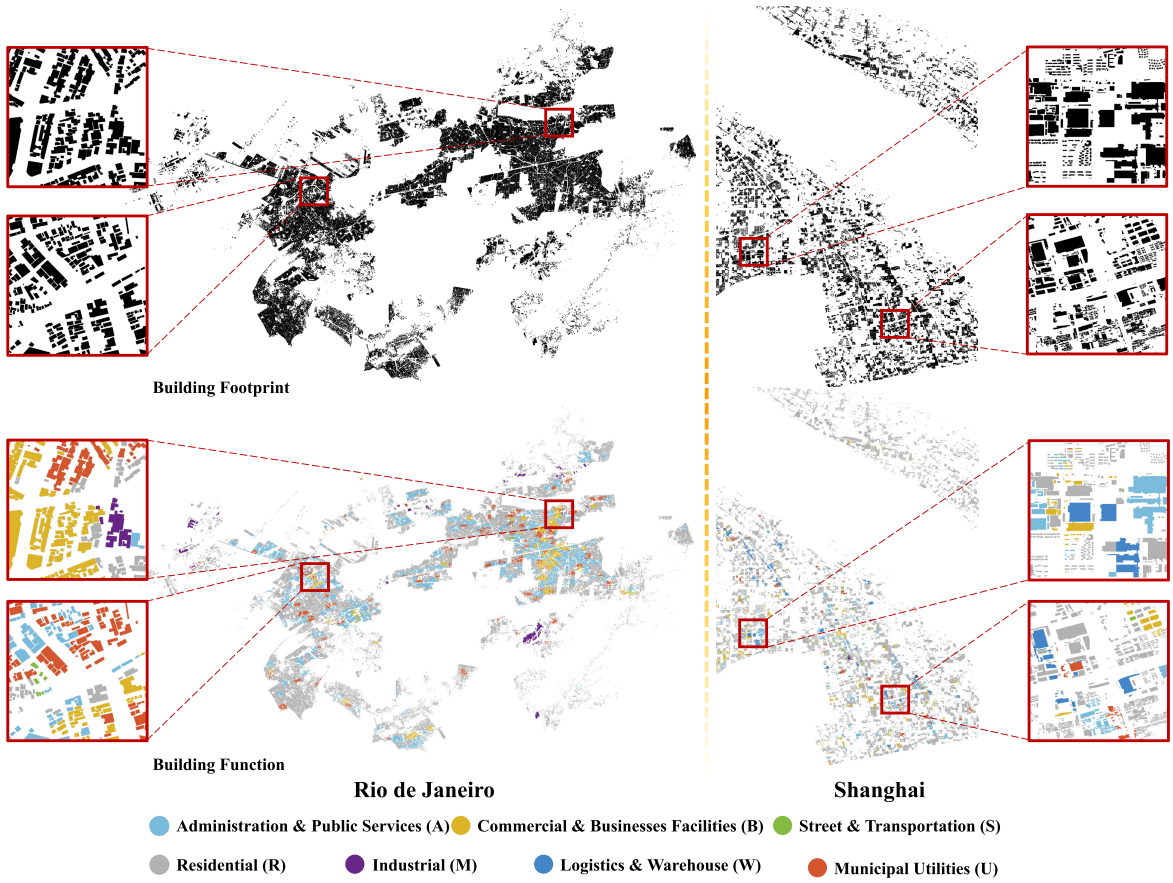


Fig. 10. Visualization of building segmentation and building function classification of Zona Oeste in Rio de Janeiro and suburban areas of Pudong New District of Shanghai, where the top row denotes the building segmentation results, and the bottom row represents the urban building function classification based on the combination of footprint and POI.

TABLE X
URBAN MORPHOLOGY AND FUNCTIONAL INDICES OF PU DONG (SHANGHAI) AND ZONA OESTE (RIO DE JANEIRO)

Indicator		Pu Dong	Zona Oeste
Morphology	MBA (m^2)	364.62	180.12
	MBH (m)	14.71	7.38
	BD	68.04	51.14
	BSI	0.025	0.098
	BSLI	24.79	24.41
Function	FXI	0.591	0.869

The data of Rio de Janeiro are collected from the DigitalGlobe WorldView 2 satellite imagery at a 0.5-m resolution, covering 2544 km^2. The data of Shanghai are collected from the DigitalGlobe WorldView 3 satellite in a continuous image strip. It contains the 0.3-m resolution Pan-Sharpened 3-band and 8-band 16-bit geotiff.

Points of Interest: We gathered and geo-coded POI data by open street map (OSM)[3] using web scraping techniques to evaluate urban developments from the functional dimension and attained POI data comprising a total of 1969 POIs from Rio de Janeiro, Brazil, and 8208 POIs from Shanghai, China. We reclassify the obtained POI data into seven land-use categories, according to the National Urban Land Use and Planning Standard (GB50137-2011). The classification criteria are shown in Table IX. Given the functional attributes of buildings, we ignore the green space in the reclassified categories and delete the corresponding data in the original data.

[3]https://www.openstreetmap.org/

D. Urban Morphology Analysis

According to the results of the urban morphology index calculation in Table X, Zona Oeste in Rio de Janeiro has a medium building scale. The moderate BD and balanced building distribution reflect that the city may have a comparatively concentrated functional area. The urban morphology dominated by medium and high-rise buildings provides residents with diverse residential and commercial choices to meet different levels of residential and office needs. A livable urban environment is established through rational open space utilization and moderate BD.

The distribution of buildings with different attributes in urban blocks reflects the development status and planning management strategy of a city (Fig. 10). We identify urban functional areas based on a combination of building footprints extracted from remote sensing images and geospatial data from crowd-sourced data. This combination breaks the limitations of statistical data and enables a holistic perception and understanding of the city. To further

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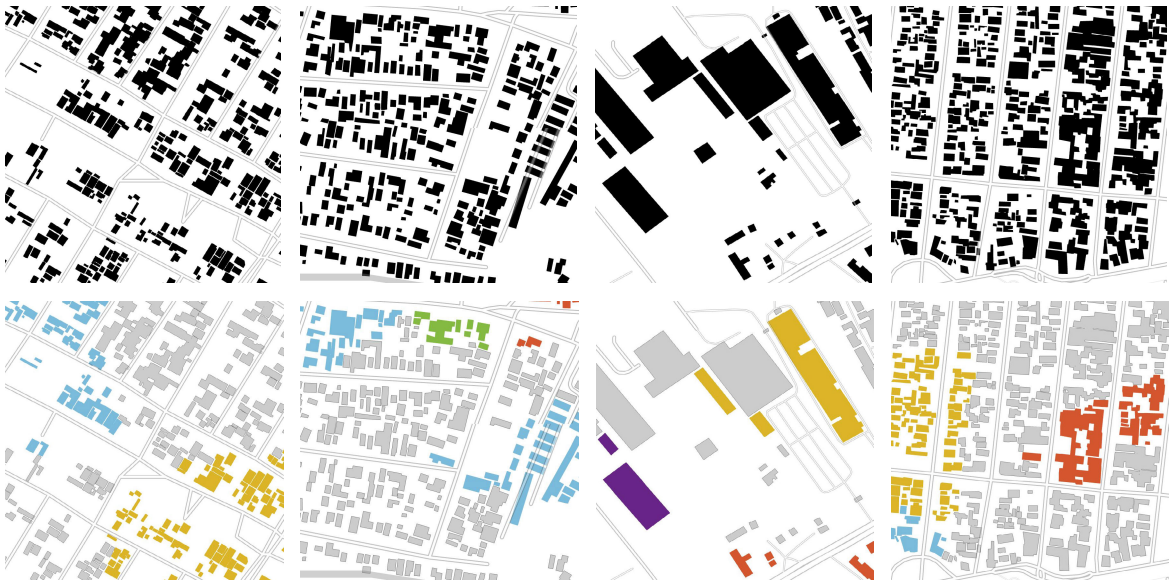


Fig. 11. Detailed visualizations of building segmentation polygons and their corresponding function classifications.

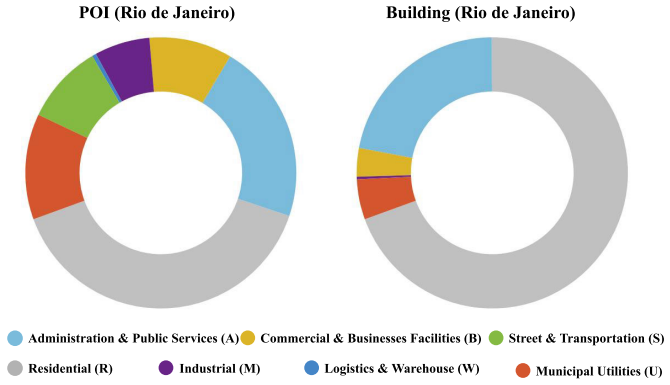


Fig. 12. Distribution of buildings with different functions and their corresponding POIs in Zona Oeste, Rio de Janeiro.

illustrate the details of building footprint segmentation and functional classification, we present a set of visual results in Fig. 11.

As shown in Fig. 12, blocks with different functions intertwine and scatter throughout the Zona Oeste, reflecting the diversity and unevenness characteristics of building functions. The residential, administration, and public services, as well as municipal utilities, occupy the majority of buildings (96.3%) in Zona Oeste, while the remaining four functions account for less than 4%.

The dense population concentrated in residential areas reflects the dominance of the urban residential function. The concentration of administrative and public service facilities represents the importance of governmental institutions, education, healthcare, and cultural and recreational facilities in urban planning. Meanwhile, the high density of municipal facilities indicates a well-developed public service system. On the contrary, the scarcity of commercial, industrial, and logistical facilities, along with the insufficient proportion of transportation and transit facilities, indicates a singular industrial structure in the urbanization process. This industrial structure is primarily dominated by service and administrative sectors, leading to underdeveloped industrial and logistical sectors. Additionally, the urban transportation infrastructure still requires further improvement to enhance the urban development potential against the backdrop of the freight transportation economy. In the future, urban development strategies serving as transportation and logistics hubs can be analyzed through the integration of railway and highway information.

We observed significant differences between the two sides of the main diagonal of the Zona Oeste, as shown in Fig. 13. The difference indicates that the level of urbanization in the region is characterized by an obvious spatial differentiation. Hence, we divide the City of Zona Oeste into two areas according to the main diagonal, with Region A denoting above the main diagonal and Region B below the main diagonal. The statistical analysis is displayed in Table XI.

The classification of urban functions reveals that residential, administrative, and public service facilities dominate in Region A, with residential buildings accounting for approximately 60% of the total buildings, and administrative and public service facilities accounting for about 23% of the total. This distribution pattern reflects the densely populated nature of the area and the centrality of the residential function, and, to some extent, indicates the sufficient provision of governmental organizations, education, and medical services. Commercial and business facilities account for about 5% of the total buildings and are located in the center of Region A. However, industrial and transportation facilities each comprise less than 1% of the total buildings, and logistics and warehousing facilities are completely absent, severely limiting the comprehensive development needs of the city.

Unlike Region A, administrative and public service facilities account for about 16%, reflecting a relative lack of governmental institutions, education, and medical services in Region B. Commercial facilities only account for 1.4%, demonstrating significantly less active commercial activities compared to

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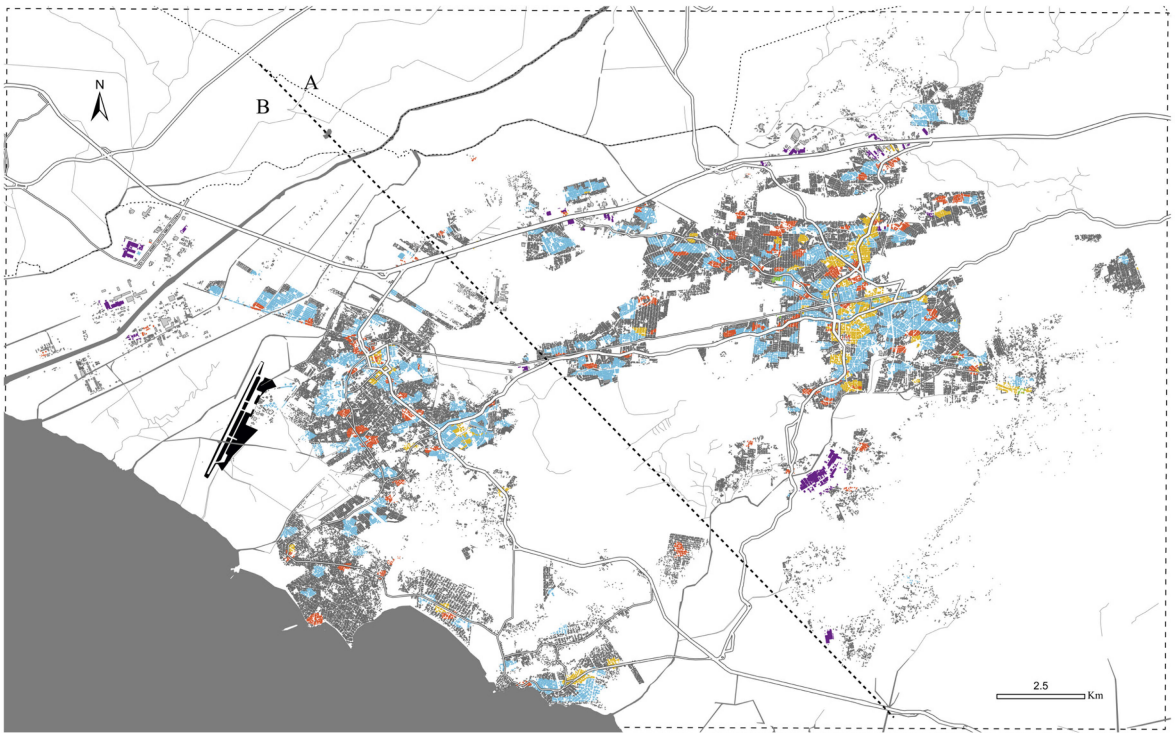


Fig. 13. Detailed urban building function of Zona Oeste in Rio de Janeiro. We divide the Zona Oeste into two different regions to support further analysis.

TABLE XI
DIFFERENCE OF THE URBAN BUILDING FUNCTION ABOUT REGIONS A AND B IN ZONA OESTE

Function	Region A	Region B
R	47,423	48,261
A	18,096	12,160
B	3,584	1,061
M	303	92
W	0	2
S	83	13
U	3,696	2,969

Region A. The approximately 4% of municipal utilities show a relatively imperfect public service system in Region B. It is worth noting that Region B contains a large amount of vacant land, providing potential space for future urban development but also indicating current underdevelopment and underutilization in the area.

E. Urban Vitality Analysis

According to Table X, urban morphology index and the function index of Zona Oeste reflect the diversity, rationality, and a high degree of efficiency in urban space utilization. The diversity of function blocks, the density of medium and high-rise buildings, and the abundance of open spaces create a vibrant urban environment. A well-organized urban plan ensures a balanced development of the functional blocks and provides a variety of options for different levels of commercial, cultural, and residential demands. By continuing to develop and improve the transportation infrastructure, the city is able to be further energized to ensure the continued economic growth of the Zona Oeste. Considering that the transfer of freight is an important source of economic activity for Zona Oeste, we recommend that the continued development of transportation and improvements to the transportation infrastructure can further stimulate urban vitality and ensure the continued economic growth of the City of Zona Oeste.

F. Conclusion

We introduce a constructive framework named CityInsight for urban analysis. We couple geo-tagged data with Earth observation data by integrating building footprints from remote sensing images with POIs to achieve recognition of building functional categories. More importantly, this strategy provides new avenues for subsequent urban analysis and data support. We primarily assessed the morphology, functional diversity of the city, and their application in urban vitality analysis, achieving novel, large-scale urban insights and perceptions based on artificial intelligence. This is our initial attempt. Despite facing many limitations and challenges, we will strive to expand this research endeavor. In the future, we dedicate ourselves to gaining a deeper understanding and insights into urban analysis from a more comprehensive perspective by integrating 3-D spatial data, and aim to make further contributions to the remote sensing community.

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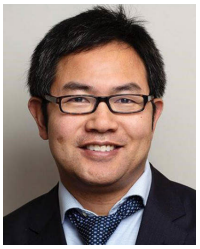
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